

Security Operations Center - Threat Analysis Report

Threat Level: MEDIUM | **Total Alerts:** 3 **Generated:**
2025-11-05T21:48:37.644Z

Executive Summary

Three critical alerts were detected targeting the Linux kernel version 5.15.0-161-generic, with vulnerabilities CVE-2021-3773, CVE-2024-56180, and CVE-2025-27558. All alerts originated from the Suricata NIDS sensor (192.168.56.104), which is classified as internal infrastructure. No external threat sources, inbound or outbound activity, or lateral movement indicators were observed. The alerts indicate potential exploitation of unpatched kernel vulnerabilities on a monitored host, but no evidence of active compromise or external connection. The absence of external IPs or malicious payloads suggests these are system-level vulnerability detections rather than active intrusion attempts. Immediate patching is recommended to prevent exploitation.

Key Findings

- All three alerts are critical severity (rule_level 13) and relate to unpatched kernel vulnerabilities in linux-image-5.15.0-161-generic.
- Alerts originated from the Suricata NIDS sensor (192.168.56.104), confirmed as internal infrastructure.
- No external IPs, C2 indicators, or data exfiltration patterns detected.
- Threat direction is unknown, but no outbound or lateral movement activity observed.
- No geolocation data available for infrastructure IP; alerts are system-level vulnerability detections.

Top Priority Threats

IP Address	Type	Country	Direction	Activity	Severity	Confidence	Count
192.168.56.104	External	N/A	Inbound	Kernel vulnerability detection	CRITICAL	High	1

MITRE ATT&CK Mapping

- **T1078.004** - Cloud Accounts (Defense Evasion)
- **T1190** - Exploit Public-Facing Application (Initial Access)
- **T1589.001** - Credentials (Reconnaissance)

Immediate Actions Required

1. Patch the linux-image-5.15.0-161-generic kernel on the Suricata NIDS sensor immediately.
2. Verify system integrity and ensure no unauthorized kernel module loading.
3. Review patch management policies for critical kernel updates.
4. Confirm that the sensor is not running as a public-facing service.
5. Monitor for subsequent alerts related to kernel exploitation or privilege escalation.

Analysis Complete Report generated: 2025-11-06 05:48:52 Threat level: MEDIUM Priority actions: 0 identified